

Stakeholder Question Catalogue 01

Undergraduate Students

80 undergraduate questions mapped to feasible sensing, authoritative building data, analysis methods, uncertainty and safe answer boundaries.

STAKEHOLDER	Undergraduate students using Abacws for classes, independent study, group work, projects, examinations, events and building services.
QUESTION RANGE	Immediate status, historical comparison, diagnosis, prediction, recommendation, optimisation, navigation, sustainability, fault reporting and evidence transparency.
EVIDENCE MODEL	Proposed room-level sensing combined with authorised occupancy, BMS, energy, network, timetable, booking, spatial, maintenance, accessibility, weather and service-status data.

Core principle: help the student decide while stating what is observed, looked up, calculated, inferred, forecast, recommended, restricted or not assessable.

1. Catalogue scope and design principles

The catalogue focuses on decisions that undergraduate students may reasonably want a conversational university building to support.

Proposed capability, not current fact

Every sensor, data feed and service in this catalogue is a design requirement unless its current deployment, coverage, permissions and operational status have been verified.

Current building information

The May 2021 floor plan provides historical room identifiers, capacities and spatial examples. Current room use, student access, bookability, entrances, lifts, services and HVAC zones require verification.

Question-to-evidence design

Each answer begins with the student decision, resolves place and time, identifies the minimum evidence, checks source quality and authority, performs the required analysis, and states uncertainty.

Correct spatial and temporal scope

A room claim requires evidence from that room or a validated served zone and the requested period. Corridor readings and stale observations may be context but not substitutes.

Authoritative source precedence

Bookings, access, lift status, closures, network, AV, maintenance and emergencies must come from authorised operational systems; environmental sensors cannot infer these states.

Data-quality gating

Location, calibration or commissioning, timestamp, uptime, missingness, drift, coverage and source disagreement are checked before any comparison, forecast, diagnosis or recommendation.

Privacy, accessibility and permissions

Use anonymous room-level occupancy and sound statistics rather than identities or speech. Apply student permissions and treat step-free access, lift status and temporary closures as hard constraints.

Safe uncertainty and refusal

Forecasts are probabilistic, recommendations need a fallback, and missing or conflicting critical evidence should produce a limited answer, clarification or not-assessable outcome.

2. How to read the catalogue

Readiness describes the verified sensing, authoritative integrations, validation and governance needed to answer responsibly; it does not rank student value.

R1 - practical with core feeds

Plausible once core room sensing and basic authoritative spatial, access, booking or closure feeds are verified and quality-controlled.

R2 - integration-dependent

Requires additional authoritative integrations such as live timetable, booking, network/AV, lift, BMS, maintenance or calibrated forecasting services.

R3 - research-grade or restricted

Requires dense or specialist sensing, validated prediction/diagnosis, restricted operational data, or tightly bounded privacy, safety and security governance.

Complexity

- L1 Direct status, location, capability or permission lookup.
- L2 Comparison, historical aggregation or bounded recommendation.
- L3 Forecast, ranking or route recommendation with uncertainty.
- L4 Diagnostic, optimisation, evidence-quality or intervention analysis.

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GLOBAL ANSWER RULES

Use only verified student-accessible candidates; preserve source timestamps and permissions; use anonymous occupancy and numerical sound statistics; distinguish observation, lookup, calculation, inference, forecast and recommendation; never substitute corridor data for a room; and return a limited or not-assessable answer when critical evidence is missing.

3. Worked example: future quiet-study recommendation

The building must resolve the requested time and destination, restrict candidates by current access and use, verify evidence quality, forecast conditions and explain uncertainty rather than returning a single unqualified location.

Student: "I am planning to go to Abacws next Wednesday after 2 p.m. What would be the ideal place to sit - a calm and relatively quiet place with less foot traffic?"

1. Resolve the request

Resolve the actual Wednesday, 90-minute window, 4 p.m. destination, meaning of quiet, and any optional power, Wi-Fi or step-free requirements.

4. Forecast the time window

Combine comparable weekdays with timetable, bookings, events and weather; produce candidate-specific probabilities and confidence intervals rather than a seat guarantee.

2. Build authorised candidates

Use the current space registry, access and booking rules; exclude closed, reserved, staff-only or incompatible spaces before considering sensor data.

5. Rank transparently

Apply hard constraints first, then rank quietness, foot traffic, environmental stability, walking time and forecast robustness; retain a viable backup.

3. Verify evidence quality

Require room-level occupancy and sound coverage, source timestamps and health status; use environmental measurements only where they are correctly located and current.

6. Communicate answerability

State the best option, backup, evidence period, confidence, important missing data and refresh time; omit or withhold any criterion that cannot be supported.

ILLUSTRATIVE ANSWER STRUCTURE

"The strongest forecast is [verified space], mainly because it is expected to have lower occupancy and sound during your 90-minute window while remaining close to [destination]. [Backup space] is the next-best option. Confidence is [low/moderate/high] because [data coverage and timetable uncertainty]. This is a room-level forecast, not a guaranteed seat; please recheck around [time]. [Criterion] was not used because [missing or stale source]."

1. Planning a place to study

Questions UG-001-UG-002

UG-001

CORE

L3 - prediction and recommendation

R2 - integration-dependent

Student question: I'm coming to Abacws next Wednesday after 2 p.m. for about 90 minutes. Where am I most likely to find a calm, relatively quiet place with low foot traffic and reasonable measured temperature and air-quality indicators, close to my 4 p.m. class?

Sensors and telemetry

Room-level air temperature, RH, direct NDIR CO₂, PM_{2.5} where deployed, task-area illuminance and privacy-preserving LAeq/L10/L90 sound statistics; anonymous room occupancy and corridor footfall. Use time-synchronised data with last update, device health and candidate-space coverage. No raw audio or speech content may be retained.

Supporting building data

Authoritative: current verified student-accessible spaces, bookings, timetable, room capacity and permitted use, opening hours, route graph, next-class location and weather forecast. Use only timetable details the student chooses to share.

Analysis required

Forecast and recommendation: resolve the date, duration and destination; forecast occupancy, footfall, sound and measured room conditions from comparable Wednesdays plus scheduled activity; exclude inaccessible, closed or reserved spaces; rank a primary and backup option and return confidence, evidence coverage and refresh time.

Answer boundary

This is a room-level forecast, not a guaranteed seat or a universal comfort judgement. Ask for missing hard constraints such as power or step-free access, and do not rank any option whose access, booking status or essential evidence is unverified.

Why this is useful: Helps the student choose where and when to study before travelling, while providing a practical backup if conditions change.

UG-002

CORE

L3 - prediction and recommendation

R2 - integration-dependent

Student question: Which two-hour periods next week are most likely to be quiet enough for focused study in Abacws?

Sensors and telemetry

Historical room-level LAeq/L10/L90 sound statistics, anonymous occupancy, corridor footfall, air temperature and illuminance for monitored study spaces, with coverage, uptime and last-calibration metadata.

Supporting building data

Authoritative: timetable, bookings, events, opening hours and verified study-space access.

Analysis required

Forecast and recommendation: evaluate only verified study spaces and two-hour windows; compare typical and upper-percentile sound, crowding and condition variability for equivalent weekdays; return several robust options, the number of comparable weeks and forecast uncertainty.

Answer boundary

Sound statistics cannot reveal speech content or guarantee silence. Closed or rarely used spaces must not appear artificially quiet, and unmonitored spaces must be labelled unknown rather than ranked as poor.

Why this is useful: Helps a student plan a repeatable study routine around conditions that are usually suitable.

1. Planning a place to study

Questions UG-003-UG-004

UG-003

CORE

L4 - itinerary optimisation

R2 - integration-dependent

Student question: I have classes on Levels 1, 3 and 5 tomorrow. Where should I study between them so I spend less time walking and searching for a place?

Sensors and telemetry

Anonymous room occupancy, corridor and stair footfall, room sound statistics and measured environmental conditions; authoritative lift and door-status telemetry only when a step-free route is requested. No raw audio or speech content may be retained.

Supporting building data

Authoritative: personal timetable supplied with consent, student-accessible spaces, bookings, route graph and room capacities.

Analysis required

Itinerary optimisation: model each timetable gap separately; include walking time, class-change congestion, forecast space availability and likely condition stability; minimise unnecessary movement while preserving enough transfer time and provide a fallback for each gap.

Answer boundary

Use only timetable entries and current location information supplied with consent. Do not infer attendance or continuously track the student, and do not promise that a forecast space will remain available.

Why this is useful: Reduces transition time and makes a full day in a six-storey building easier to manage.

UG-004

CORE

L3 - prediction and recommendation

R2 - integration-dependent

Student question: Where can I work for three hours on Friday with power, good Wi-Fi, useful daylight and a low risk of overheating?

Sensors and telemetry

Room air temperature, task-area illuminance and anonymous occupancy; aggregate Wi-Fi access-point health and room electrical-service/fault telemetry where authorised. Power-point count is an asset record, not a sensor reading.

Supporting building data

Authoritative: student-accessible-space registry, documented power points, room bookings, opening/access rules, aggregate eduroam service status, route graph and weather forecast.

Analysis required

Forecast and recommendation: filter by verified access, bookability, power provision and required services; forecast thermal and daylight conditions for the three-hour window; assess aggregate network health; rank a primary and backup option with criterion-specific confidence.

Answer boundary

Do not guarantee a free or working socket, device-specific Wi-Fi performance or personal thermal comfort. If room features, service status or sensor coverage are unknown, omit that criterion or return not assessable.

Why this is useful: Useful for long study sessions, coding work and online activity where service interruptions are costly.

1. Planning a place to study

Questions UG-005-UG-006

UG-005

CORE

L3 - fallback recommendation

R2 - integration-dependent

Student question: My preferred breakout area is often full after lunch. What is the best nearby alternative, and how confident are you?

Sensors and telemetry

Anonymous room occupancy estimate, doorway people counts, corridor footfall, privacy-preserving sound-level statistics (LAeq/L10/L90; no raw audio), room air temperature, RH and NDIR CO2. Require last update, device health and candidate-space coverage.

Supporting building data

Authoritative: verified candidate spaces, bookings, timetable, route graph and historical forecast error.

Why this is useful: Provides a useful fallback instead of sending the student to search the building manually.

Analysis required

Fallback forecast: estimate when the preferred area is likely to approach practical capacity; identify nearby spaces with the same permitted use; compare predicted occupancy, sound and measured conditions; return the best alternative, second choice and forecast confidence.

Answer boundary

Use anonymous room-level counts, not identities. A low occupancy estimate is not proof of a free seat, and the answer should advise rechecking shortly before arrival when confidence is low.

UG-006

SITUATIONAL

L3 - long-horizon recommendation

R2 - integration-dependent

Student question: I will be in Abacws from 9 a.m. to 5 p.m. Which study area is likely to work best as a base for the day, and when should I expect to need a backup?

Sensors and telemetry

Historical anonymous room occupancy, privacy-preserving sound-level statistics (LAeq/L10/L90; no raw audio), air temperature, RH, NDIR CO2, PM2.5 where monitored and illuminance. Require last update, health and room coverage.

Supporting building data

Authoritative: timetable, bookings, planned events, opening hours and space-use rules.

Why this is useful: Helps commuters and students with long campus days avoid repeated relocation.

Analysis required

Long-horizon forecast: compare candidate spaces across the full day, penalising predictable class-change peaks, closures and afternoon thermal or glare risk; show likely change points and pre-plan a nearby backup rather than returning one eight-hour guarantee.

Answer boundary

A full-day forecast is inherently less certain than a current-status answer. It must show time-varying confidence, never assume a seat will remain occupied for the student, and refresh when timetable or live conditions change.

1. Planning a place to study

Questions UG-007-UG-008

UG-007

SITUATIONAL

L4 - multi-objective optimisation

R3 - research-grade

Student question: I need a quiet hour, then an hour of group work, and then my lecture. Can you suggest the best sequence of spaces and when I should move?

Sensors and telemetry

Anonymous room occupancy, doorway and corridor footfall, privacy-preserving sound-level statistics (no raw audio), room temperature and NDIR CO2 forecasts, and authorised lift/door status where routing requires them. Require source health and spatial coverage.

Supporting building data

Authoritative: timetable, candidate-space rules, bookings, room capacity and the student's stated preferences.

Analysis required

Multi-objective itinerary optimisation: represent quiet study, discussion and lecture attendance as different constraint sets; optimise space sequence and departure times using forecast occupancy, sound, conditions and walking risk; re-plan only when live evidence materially changes.

Answer boundary

Optimise building-related choices only. Do not read private calendar content, create bookings or store accessibility preferences without explicit consent; present alternatives rather than directing the student to a restricted space.

Why this is useful: Supports a realistic day with different study modes and reduces unnecessary movement between floors.

UG-008

CORE

L3 - prediction and recommendation

R2 - integration-dependent

Student question: Where can two of us revise together next Tuesday without disturbing quiet-study users or sitting beside a busy corridor?

Sensors and telemetry

Room and nearby-corridor LAeq/L10/L90 sound statistics, anonymous occupancy and footfall, plus room air temperature and direct NDIR CO2 where monitored. Sound processing must remain numerical and local.

Supporting building data

Authoritative: space-use classification, student access, bookings and timetable.

Analysis required

Forecast and recommendation: restrict candidates to spaces where quiet conversation is permitted; forecast occupancy, background sound and corridor spill-over for the requested time; rank by suitability for two people and walking distance.

Answer boundary

Do not record speech or treat any empty space as discussion space. Current use rules and bookings override occupancy sensors, and unverified access or acoustic coverage must be stated.

Why this is useful: Supports group learning while protecting the experience of students seeking quiet study.

1. Planning a place to study

Questions UG-009-UG-010

UG-009

CORE

L2 - availability comparison

R2 - integration-dependent

Student question: Which student study areas are officially open after 5 p.m. this week, and which are likely to have usable space and acceptable measured conditions?

Sensors and telemetry

Historical anonymous room occupancy and room air temperature and direct NDIR CO2 trends, plus authorised door, closure and building-service status where integrated. Require source timestamp, health and spatial coverage.

Supporting building data

Authoritative: opening hours, student-card access rules, planned closures, bookings and maintenance notices.

Analysis required

Verified lookup plus forecast: first return spaces officially open and accessible after 5 p.m.; then estimate likely crowding and measured condition stability for those spaces only. Keep the official opening/access result separate from the forecast.

Answer boundary

Access cannot be inferred from a door contact, occupancy estimate or old floor plan. If an authoritative opening, closure or card-access feed is unavailable, say that access is unconfirmed.

Why this is useful: Useful for students studying after teaching hours or around employment and caring responsibilities.

UG-010

CORE

L4 - consequence-sensitive recommendation

R2 - integration-dependent

Student question: I have an online interview next week. Which bookable room and time give me the lowest risk of noise, interruptions, overheating, power problems or poor Wi-Fi?

Sensors and telemetry

Room LAeq/L10/L90 sound statistics, anonymous occupancy, air temperature and direct NDIR CO2; aggregate access-point health, power/AV service status and door-event telemetry from authorised systems. No ambient speech recording. No raw audio or speech content may be retained.

Supporting building data

Authoritative: authoritative bookable-room inventory, privacy/use rules, bookings, aggregate eduroam service status, maintenance/closure notices and timetable context.

Analysis required

Consequence-sensitive forecast and recommendation: require a bookable room with appropriate privacy, power and network service; forecast interruption, sound, crowding and measured room conditions; return a primary slot, backup and separate risk explanations.

Answer boundary

An open breakout area is not a private interview room. Network, power and environmental predictions are not guarantees, and missing authoritative booking or service data makes the recommendation not assessable.

Why this is useful: A failed interview has high personal cost, making this a strong student-centred use case.

2. Live study-space selection and movement

Questions UG-011-UG-012

UG-011

CORE

L2 - real-time recommendation

R1 - practical with core feeds

Student question: I'm at the main entrance now. What is the quietest student-accessible study area I can reach within five minutes?

Sensors and telemetry

Current room sound, anonymous occupancy and corridor footfall. Use current, time-synchronised room/zone data and aggregate counts only. No raw audio or speech content may be retained.

Supporting building data

Authoritative: current student-accessible spaces, permitted use, opening/closure status, route graph and lift/door state. Use the student's current location only for this request unless they opt in to longer use.

Why this is useful: Avoids wandering between floors when a student needs a place immediately.

Analysis required

Current observation and recommendation: identify spaces reachable within five minutes, exclude closed, reserved or discussion-only areas, compare current sound and estimated spare capacity, and return the best verified option with a nearby fallback.

Answer boundary

Recommend a room or area, not a specific free seat unless validated seat-level sensing exists. Unmonitored spaces are unknown, and stale or indirect data require a qualified answer or no recommendation.

UG-012

CORE

L2 - real-time comparison

R1 - practical with core feeds

Student question: For one hour of quiet work right now, which student-accessible study area is the strongest option?

Sensors and telemetry

Current anonymous room occupancy, privacy-preserving sound-level statistics (LAeq/L10/L90; no raw audio), air temperature and RH, NDIR CO2, PM2.5 where monitored, and illuminance; use time-synchronised room-level data.

Supporting building data

Authoritative: verified access and space-use rules; route graph.

Why this is useful: Provides a transparent comparison rather than presenting the nearest sensor reading as a recommendation.

Analysis required

Current comparison and recommendation: compare only verified, discussion-compatible spaces; normalise occupancy by practical capacity, examine current sound and measured conditions, and explain why the recommended option outranks the alternatives.

Answer boundary

Ask which criteria matter when "best" is underspecified. Do not convert a low room count into a seat guarantee or call measured temperature and CO2 a complete comfort assessment.

2. Live study-space selection and movement

Questions UG-013-UG-014

UG-013

CORE

L3 - short-horizon forecast

R2 - integration-dependent

Student question: Which area has just become less crowded and is likely to stay that way for the next 45 minutes?

Sensors and telemetry

Live room occupancy, doorway counts and corridor footfall. Use current, time-synchronised room/zone data and aggregate counts only.

Supporting building data

Authoritative: nearby class end times, bookings and historical arrival patterns.

Analysis required

Short-horizon forecast: detect recent departures from occupancy and doorway-count changes; incorporate incoming bookings and nearby class end times; rank spaces by the probability of retaining spare capacity for the full 45-minute period.

Answer boundary

A temporarily empty area may have an imminent booking or class-change influx. Use aggregate counts only and give probability and refresh time, not a certainty.

Why this is useful: Prevents a misleading recommendation based on a momentary low count.

UG-014

SITUATIONAL

L2 - real-time recommendation

R3 - validated daylight model required

Student question: Where can I sit now with useful daylight but a low risk of glare or solar overheating?

Sensors and telemetry

Task-plane illuminance, vertical/window-zone illuminance or a validated luminance proxy, outdoor solar irradiance, blind position and air/radiant temperature by seating zone; anonymous occupancy for candidate availability.

Supporting building data

Authoritative: current room orientation, window and blind geometry, verified student access, current weather and usable seating zones.

Analysis required

Current assessment and recommendation: estimate daylight availability, glare risk and solar-heat exposure by zone using measured light plus geometry; rank valid seating areas and show how the result may change as the sun moves.

Answer boundary

Illuminance alone cannot establish glare or individual visual comfort. Without validated geometry and suitably placed light sensors, report only measured brightness and do not make a zone-level glare recommendation.

Why this is useful: Supports visual comfort while making sensible use of daylight.

2. Live study-space selection and movement

Questions UG-015-UG-016

UG-015

CORE

L2 - personalised comparison

R1 - practical with core feeds

Student question: I feel cold today. Which student-accessible study area is warmer without showing clearly worse CO2 or particle levels?

Sensors and telemetry

Room air temperature and RH, direct NDIR CO2, humidity-corrected PM2.5 where deployed, anonymous occupancy and sensor-health metadata. TVOC may be shown only as an indicative supplementary signal.

Supporting building data

Authoritative: student-accessible spaces and route distance.

Analysis required

Current observation and preference-based recommendation: compare verified spaces using the student's stated preference; rank warmer options only when CO2 and PM2.5 are not materially worse and explain the trade-off.

Answer boundary

Do not make a medical or universal comfort claim. CO2 is a ventilation indicator, PM sensors require quality control, and a warmer room may still be unsuitable for other reasons.

Why this is useful: Responds to an immediate preference while avoiding a simplistic "warmest is best" answer.

UG-016

SITUATIONAL

L2 - environmental recommendation

R2 - integration-dependent

Student question: I want to avoid high particle levels today. Which monitored student study areas currently have the lowest trustworthy PM2.5 readings?

Sensors and telemetry

Indoor and outdoor PM2.5/PM10 sensors with humidity correction, calibration or co-location record, last update and health state; room occupancy and ventilation state as context. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: student-accessible spaces and current room availability.

Analysis required

Current observation and comparison: quality-check the particle sensors, compare only adequately monitored student spaces, distinguish likely outdoor influence where possible, and label unmonitored spaces unknown rather than low.

Answer boundary

This supports an environmental preference, not medical advice or a declaration of safety for an individual condition. If calibration, placement or freshness is inadequate, do not rank the space.

Why this is useful: Provides a bounded, evidence-based choice for students sensitive to particulate conditions.

2. Live study-space selection and movement

Questions UG-017-UG-018

UG-017

CORE

L3 - dynamic accessible routing

R2 - integration-dependent

Student question: What is the least crowded step-free route from my current location to the recommended study area?

Sensors and telemetry

Corridor and stair footfall, lift and door status. Use current, time-synchronised room/zone data and aggregate counts only.

Supporting building data

Authoritative: accessible route graph, temporary closures and current location supplied with consent.

Analysis required

Dynamic accessible-route recommendation: remove stairs, restricted paths and unavailable lifts/doors; estimate short-horizon route congestion from aggregate footfall; return the least-crowded verified route, travel-time trade-off and fallback.

Answer boundary

Use a location supplied for this request rather than continuous tracking. Accessibility and lift/door status must be authoritative; stale status makes the route unverified.

Why this is useful: Combines space recommendation with inclusive navigation.

UG-018

SITUATIONAL

L4 - evidence-quality query

R3 - research-grade

Student question: The app says this area is nearly empty. How reliable is that estimate, and are any occupancy sensors or data feeds unavailable?

Sensors and telemetry

At least two independent aggregate occupancy sources where feasible, such as a doorway counter and room presence/count sensor, plus device uptime, last update, count-reconciliation and fault state.

Supporting building data

Authoritative: room capacity, bookings, timetable and sensor commissioning/maintenance records; no personal identity or device identifiers.

Analysis required

Evidence-quality assessment: compare independent occupancy sources, detect stale updates and doorway-counter drift, reconcile with booking/timetable context, and return a plausible occupancy interval with confidence and missing-source flags.

Answer boundary

Do not infer identities, attendance or a guaranteed free seat. Material disagreement between independent sources should produce an uncertain or not-assessable occupancy result.

Why this is useful: Helps students calibrate trust in a recommendation and exposes sensor limitations.

2. Live study-space selection and movement

Questions UG-019-UG-020

UG-019

CORE

L3 - move-or-stay forecast

R2 - integration-dependent

Student question: Should I stay here, or are noise, crowding or CO2 likely to get noticeably worse during the next hour?

Sensors and telemetry

Current privacy-preserving sound statistics, anonymous room occupancy, corridor footfall, NDIR CO2 and air-temperature trends; use time-synchronised sources and current health status. No raw audio or speech content may be retained.

Supporting building data

Authoritative: upcoming bookings, nearby class changes and comparable historical periods.

Analysis required

Move-or-stay forecast and recommendation: define what counts as a noticeable change, forecast sound, occupancy and CO2 for the next hour, compare the expected disruption with the cost of moving, and provide a nearby fallback.

Answer boundary

Explain which factor drives the advice. CO2 is not a complete air-quality measure, and the recommendation must be withheld when current room coverage or upcoming booking data are inadequate.

Why this is useful: Supports a practical decision before conditions become disruptive.

UG-020

CORE

L3 - contingency recommendation

R2 - integration-dependent

Student question: If this area fills up, which nearby alternative has the most similar quietness, lighting and environmental conditions?

Sensors and telemetry

Current privacy-preserving sound statistics, illuminance, anonymous room occupancy, air temperature, RH and NDIR CO2; use time-synchronised room-level data and source health. No raw audio or speech content may be retained.

Supporting building data

Authoritative: route graph, verified alternatives, capacity and access rules.

Analysis required

Contingency recommendation: identify nearby student-accessible spaces with a similar measured profile, forecast their occupancy and conditions, and state the trigger at which moving becomes worthwhile.

Answer boundary

Explain similarity by criterion rather than hiding it in one score. Do not imply a free seat or current access when booking, occupancy or route data are unverified.

Why this is useful: Gives the student a ready fallback instead of forcing a new search.

3. Lectures, classes and exams

Questions UG-021-UG-022

UG-021

SITUATIONAL

L3 - zone recommendation

R3 - dense room sensing required

Student question: For a lecture scheduled in Room 2.26, is the front, middle or rear usually less stuffy and less noisy?

Sensors and telemetry

Multiple occupied-zone NDIR CO2, temperature, RH and privacy-preserving sound sensors distributed front-middle-rear inside the theatre; aggregate occupancy distribution and supply-airflow telemetry. No raw audio or speech content may be retained.

Supporting building data

Authoritative: current room use verified against authoritative timetable, expected attendance, room geometry, sensor positions and HVAC service relationships.

Analysis required

Historical zone comparison: match classes by duration and occupancy, compare front/middle/rear CO2, temperature and sound distributions, and return only a zone-level tendency with uncertainty and the number of comparable sessions.

Answer boundary

A single room-average or corridor sensor cannot answer this. If the current room use differs from the May 2021 plan or multi-point occupied-zone coverage is absent, return not assessable.

Why this is useful: Large lecture theatres can be environmentally uneven, so a room-wide average may not help a student choose where to sit.

UG-022

CORE

L3 - class-condition forecast

R2 - integration-dependent

Student question: My two-hour class is scheduled in Room 0.01 this afternoon. Is the room likely to become too warm or develop high CO2?

Sensors and telemetry

Multiple occupied-zone air-temperature, RH and NDIR CO2 sensors inside the room; aggregate occupancy count; supply-airflow, fan/damper and control-state telemetry.

Supporting building data

Authoritative: current room use, class duration and expected attendance from authoritative timetable data; recent room use, room volume/HVAC zone and weather forecast.

Analysis required

Session forecast: initialise from current multi-zone room conditions, expected attendance and HVAC state; use comparable sessions or a validated room-specific model to forecast temperature and CO2 through the two-hour class with confidence bands.

Answer boundary

This forecasts measured indicators, not safety, compliance or individual comfort. Missing attendance, room-level sensing or HVAC state requires a limited forecast or not-assessable result.

Why this is useful: Allows a student to prepare, choose a seating zone or raise a concern before a long class.

3. Lectures, classes and exams

Questions UG-023-UG-024

UG-023

SITUATIONAL

L2 - historical comparison

R2 - integration-dependent

Student question: For large classes like mine, which current teaching rooms have kept the most stable temperature, CO2 and background sound this term?

Sensors and telemetry

Inside-room air temperature, direct NDIR CO2, LAeq/L10/L90 sound statistics and anonymous occupancy, with multi-point coverage in large rooms and common timestamps. No raw audio or speech content may be retained.

Supporting building data

Authoritative: timetable, class size, session duration, room capacity and weather.

Analysis required

Matched historical comparison: select classes with similar duration, attendance and time of day; adjust for outdoor weather; compare variability, upper-percentile values and recurring episodes rather than raw averages.

Answer boundary

This is descriptive evidence for feedback, not authority to change a timetable. Do not compare unlike rooms or sessions, and return limited results when current room use or sensor coverage is unverified.

Why this is useful: Identifies recurring differences between rooms and can support student-representative feedback.

UG-024

CORE

L2 - recovery assessment

R1 - practical with core feeds

Student question: The previous class has just ended. Are CO2 and temperature back near this room's usual pre-class levels?

Sensors and telemetry

Inside-room NDIR CO2, air temperature and RH, anonymous occupancy and supply-airflow/fan state; large teaching spaces require multiple occupied-zone points with current health metadata.

Supporting building data

Authoritative: previous and next timetable slots; door state where relevant.

Analysis required

Current observation plus recovery estimate: compare present CO2 and temperature with the pre-class baseline, model the recent decay under current airflow, and state whether the room is recovering, stable or still elevated before the next class.

Answer boundary

A corridor reading cannot establish room recovery. Define "recovered" against a stated room baseline or operational criterion, and return not assessable when the room sensor is unhealthy or missing.

Why this is useful: A simple, realistic question that connects current room conditions with the class timetable.

3. Lectures, classes and exams

Questions UG-025-UG-026

UG-025

CORE

L3 - short-horizon forecast

R2 - integration-dependent

Student question: Class starts in 15 minutes. Based on the current recovery rate and ventilation state, are CO2 and temperature likely to be back near their usual pre-class levels by then?

Sensors and telemetry

Inside-room air temperature, RH and direct NDIR CO2, anonymous occupancy, supply airflow/fan state and door state; use a room-specific, time-synchronised history.

Analysis required

Short-horizon forecast: estimate temperature and CO2 decay using the room's recent trajectory, occupancy state and measured airflow; compare the predicted class-start values with explicit operational criteria and show uncertainty.

Supporting building data

Authoritative: next class size, room volume and HVAC service information.

Answer boundary

Do not provide an exact recovery time from a generic model. A room-specific history or validated mass-balance/thermal model is required; otherwise report the trend only.

Why this is useful: Provides a practical near-term answer about room readiness.

UG-026

SITUATIONAL

L2 - seating-zone recommendation

R3 - validated daylight model required

Student question: Which part of this room has the lowest glare and solar-heat risk for this afternoon's class?

Sensors and telemetry

Task-plane and vertical illuminance, outdoor solar irradiance, blind state and air/radiant temperature distributed across window and internal seating zones.

Analysis required

Zone-level current assessment and forecast: combine measured light, room geometry and sun position to estimate glare and solar-heat exposure by seating zone, then show how the preferred zone may shift during the class.

Supporting building data

Authoritative: room orientation, window locations, current weather and seating layout.

Answer boundary

Do not infer individual visual comfort from illuminance alone. Without multi-zone sensing and validated geometry, report measured brightness only and return the seating recommendation as not assessable.

Why this is useful: Useful in rooms where one facade causes strong afternoon sun.

3. Lectures, classes and exams

Questions UG-027-UG-028

UG-027

SITUATIONAL

L2 - noise forecast

R2 - integration-dependent

Student question: The corridor is noisy during class change. Is the noise likely to fall enough for us to hear clearly when the lecture begins?

Sensors and telemetry

Room and corridor LAeq/L10/L90 sound statistics, door state, corridor footfall and teaching-audio system status. Where available, use commissioned room-acoustic measures such as reverberation time; do not retain speech. No raw audio or speech content may be retained. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: class-change timetable and room acoustic characteristics.

Analysis required

Short-horizon acoustic forecast: model the decay of class-change activity and its transmission through the doorway, then estimate expected room sound or signal-to-noise conditions at lecture start.

Answer boundary

Sound level alone is not speech intelligibility. If room acoustics or teaching-audio status are unknown, answer only whether background sound is falling and avoid claiming students will hear clearly.

Why this is useful: Helps distinguish a brief class-change disturbance from a recurring teaching-space problem.

UG-028

SITUATIONAL

L4 - alternative-room comparison

R3 - operational integration required

Student question: If the university offers more than one authorised replacement room for my class, which option best balances step-free access, capacity, travel time and measured room conditions?

Sensors and telemetry

Anonymous occupancy, inside-room air temperature and RH, NDIR CO2 and privacy-preserving sound statistics; authorised lift/door status. Large alternatives require multi-point coverage. No raw audio or speech content may be retained.

Supporting building data

Authoritative: timetable change, authorised alternatives, capacity, equipment, accessibility and route graph.

Analysis required

Constrained comparison and recommendation: accept only institution-provided alternatives; treat capacity, availability, required equipment and step-free access as hard constraints; compare route time and current/forecast room conditions.

Answer boundary

The student-facing system must not reassign a class or invent alternatives. It may compare only officially authorised options and must not expose restricted operational details.

Why this is useful: Supports students when a disruption creates several possible replacement rooms.

3. Lectures, classes and exams

Questions UG-029-UG-030

UG-029

SITUATIONAL

L4 - post-event diagnosis

R3 - research-grade

Student question: Last week's class felt hot and stuffy. What did the measurements show, and which contributing factors are best supported by the evidence?

Sensors and telemetry

Inside-room air temperature and RH, NDIR CO2, anonymous occupancy, supply airflow, fan/damper state and aggregate equipment power, plus outdoor temperature/solar reference.

Supporting building data

Authoritative: exact class period, expected attendance and relevant fault/maintenance events.

Analysis required

Post-event diagnostic assessment: reconstruct the class period, compare it with matched sessions, examine the timing of occupancy, temperature, CO2, airflow, solar load and equipment load, and rank contributing hypotheses by evidence strength.

Answer boundary

Distinguish observation and association from diagnosis. Do not name a failed asset or causal mechanism unless the BMS and maintenance evidence supports it; otherwise report plausible contributors and uncertainty.

Why this is useful: Turns a student complaint into a traceable investigation rather than a one-sensor anecdote.

UG-030

CORE

L3 - exam-condition forecast

R3 - high-consequence integration required

Student question: I have an exam in this room tomorrow. Are the monitored environmental conditions and essential room services likely to remain stable for the full exam?

Sensors and telemetry

Occupied-zone temperature and RH, NDIR CO2 and privacy-preserving sound statistics; aggregate occupancy history; monitored power, ventilation and teaching-room system status where integrated. No raw audio or speech content may be retained.

Supporting building data

Authoritative: authoritative exam room, start/end time and expected attendance; planned maintenance, current fault notices, room-service status and weather forecast.

Analysis required

High-consequence readiness forecast: use the official exam duration and expected attendance, compare with equivalent full sessions, check unresolved room, power, AV, ventilation and network faults, and report separate environmental and service risks with confidence and refresh time.

Answer boundary

The system may report monitored readiness and uncertainty, but it cannot decide whether an examination proceeds or what academic mitigation applies. Missing official exam, service or room-level data must be explicit.

Why this is useful: Exam conditions have high consequences for students, making this a realistic and valuable use case.

4. Group work, computer labs and equipment

Questions UG-031-UG-032

UG-031

CORE

L3 - group-space recommendation

R2 - integration-dependent

Student question: Six of us need to work together from 3 p.m. to 5 p.m. Which student-accessible space has enough seats, permits discussion, provides power and is unlikely to become stuffy?

Sensors and telemetry

Anonymous room occupancy, privacy-preserving sound statistics, air temperature and RH, NDIR CO2 and authorised electrical/service status. Sensors cannot prove booking permission or a free seat/socket. No raw audio or speech content may be retained.

Supporting building data

Authoritative: capacity, space-use rules, sockets/features, bookings, access and expected group size.

Analysis required

Forecast and recommendation: require an authorised discussion space with sufficient capacity and documented power provision; forecast occupancy and CO2 for the two-hour period; rank valid options by proximity, measured conditions and service status.

Answer boundary

An empty teaching room is not automatically available, and an occupancy count is not a seat/socket guarantee. Booking, access and permitted use must come from authoritative systems.

Why this is useful: Supports group work while avoiding spaces that are too small, too quiet or environmentally unsuitable.

UG-032

CORE

L3 - hybrid-meeting recommendation

R2 - integration-dependent

Student question: Which student-bookable room is best for a hybrid group meeting if we need low background noise, power, good Wi-Fi and suitable temperature and ventilation indicators?

Sensors and telemetry

Privacy-preserving sound-level statistics (LAeq, L10 and L90; no raw audio), anonymous occupancy, air temperature and RH, NDIR CO2 and authorised room power/service status. Network status comes from aggregate AP management telemetry.

Supporting building data

Authoritative: student-bookable rooms, AV features, eduroam access-point telemetry and bookings.

Analysis required

Multi-service forecast and recommendation: treat bookability, AV, power and access-point health as hard constraints; forecast background sound, occupancy and measured room conditions; rank rooms by the probability that all essential services remain usable.

Answer boundary

Do not guarantee device-specific Wi-Fi or acoustic privacy from a low sound level. If booking, AV, power or network status is unverified, label that criterion unknown and do not claim the room is suitable.

Why this is useful: Hybrid meetings fail when any one of several services is poor, so a multi-source recommendation is valuable.

4. Group work, computer labs and equipment

Questions UG-033-UG-034

UG-033

SITUATIONAL

L2 - society-space recommendation

R2 - integration-dependent

Student question: Where can a ten-person student society committee meet and talk without disturbing nearby quiet-study areas?

Sensors and telemetry

Room and adjacent-zone LAeq/L10/L90 sound statistics, anonymous occupancy, door state and nearby corridor footfall; room air temperature and direct NDIR CO2 where monitored. No speech recording. No raw audio or speech content may be retained.

Supporting building data

Authoritative: student booking permissions, room capacity, use rules, opening hours and adjacent-space classification.

Analysis required

Current comparison and recommendation: restrict candidates to authorised discussion spaces; estimate likely sound spill-over using door state and adjacent-zone statistics; rank rooms by capacity, separation from quiet areas and measured conditions.

Answer boundary

Do not expose private booking details or infer availability from an empty sensor reading. If adjacent acoustic coverage or room-use classification is missing, omit the disturbance claim.

Why this is useful: Supports student activities while protecting quiet study zones.

UG-034

CORE

L2 - lab and workstation recommendation

R2 - integration-dependent

Student question: Which student computer lab is open, has a reasonable chance of a free workstation and offers the software I need?

Sensors and telemetry

Aggregate workstation-management telemetry showing available/healthy sessions, room occupancy, access-point and electrical-service health; optional anonymous seat sensors. No inspection of personal files, screens or activity.

Supporting building data

Authoritative: authoritative lab opening/access rules, current software image/inventory, timetable, workstation maintenance/outage state, room capacity and route graph.

Analysis required

Authoritative service lookup and bounded recommendation: filter laboratories by current student access and required software image, estimate available healthy workstations, check outages and rank valid labs by walking distance and confidence.

Answer boundary

Do not identify logged-in users or guarantee a particular workstation. Stale software inventory or workstation telemetry must be shown as unverified rather than available.

Why this is useful: Highly relevant to computing and mathematics students who need specialist software and workstations.

4. Group work, computer labs and equipment

Questions UG-035-UG-036

UG-035

CORE

L3 - group-size forecast

R2 - integration-dependent

Student question: If eight of us use this room for two hours, how likely is CO2 to rise substantially, and is another authorised room less likely to have that problem?

Sensors and telemetry

Inside-room NDIR CO2, air temperature and RH, anonymous occupancy and supply-airflow history, with room volume and sensor health sufficient for a validated room-specific forecast.

Analysis required

Room-specific forecast and comparison: model CO2 accumulation for eight people over two hours using room volume, recent airflow and comparable sessions; compare only authorised alternative rooms and state assumptions and prediction intervals.

Supporting building data

Authoritative: room volume, capacity, ventilation relationship, booking permission and proposed time.

Answer boundary

CO2 is a ventilation indicator, not a complete air-quality assessment. Do not reuse one generic room model or recommend an alternative whose booking or ventilation evidence is unverified.

Why this is useful: Helps groups choose a room that matches their size and duration.

UG-036

CORE

L2 - project-space recommendation

R2 - integration-dependent

Student question: We will use six laptops for a project session. Which authorised room has enough power points and is least likely to become uncomfortably warm?

Sensors and telemetry

Room air temperature, anonymous occupancy and aggregate plug-load or circuit telemetry where approved; room power-point provision and fault state come from authoritative asset/service records, not inference from laptops.

Analysis required

Recommendation: filter by verified permission, capacity and power provision; use comparable occupied sessions and, where available, aggregate plug-load history to forecast thermal stability; provide a primary and backup room.

Supporting building data

Authoritative: room capacity, documented power-point provision, student booking/access rules, timetable and current electrical/service fault notices.

Answer boundary

Six laptops do not by themselves imply overload. Do not claim a particular socket works or that a circuit is safe unless approved electrical-service data supports it.

Why this is useful: Helps a project group find a permitted room with the required facilities without making unsupported electrical claims.

4. Group work, computer labs and equipment

Questions UG-037-UG-038

UG-037

SITUATIONAL

L3 - weekly scheduling

R2 - integration-dependent

Student question: We have three possible times for project meetings this week. Which time and student-bookable room combination is most likely to work well?

Sensors and telemetry

Historical anonymous occupancy, privacy-preserving sound statistics, air temperature, RH and NDIR CO2 for candidate rooms, with source coverage and health metadata. No raw audio or speech content may be retained.

Supporting building data

Authoritative: candidate times supplied by the group, student-bookable inventory, bookings, capacity, access and route graph.

Analysis required

Room-time forecast and recommendation: evaluate only the three times supplied by the group; combine bookings, expected occupancy, sound and measured conditions; rank valid room-time pairs and provide a fallback for each.

Answer boundary

Do not read group calendars, contact members or create a booking without explicit confirmation. If a room's booking, access or sensor status is unknown, exclude it or label it unverified.

Why this is useful: Supports realistic group planning while keeping personal scheduling data under user control.

UG-038

SITUATIONAL

L3 - recording-space recommendation

R2 - integration-dependent

Student question: Where can we record a presentation with low background noise, reliable power and a low risk of interruptions?

Sensors and telemetry

Room and nearby-zone privacy-preserving sound-level statistics (LAeq/L10/L90; no raw audio), anonymous occupancy, door events and authorised power/AV status; no ambient speech recording.

Supporting building data

Authoritative: student-bookable rooms, room-use rules, bookings, AV features and access.

Analysis required

Forecast and recommendation: require a student-bookable room with suitable use rules, power and any needed AV; forecast room and nearby-zone sound plus interruption risk from bookings and door activity; return a backup.

Answer boundary

Low measured sound does not guarantee acoustic isolation or permission to record. Never capture ambient speech, and do not claim power/AV availability without an authoritative service status.

Why this is useful: Useful for assessed presentations, demonstrations and project videos.

4. Group work, computer labs and equipment

Questions UG-039-UG-040

UG-039

CORE

L1 - permission and availability query

R1 - practical with booking feed

Student question: This room is booked but looks empty. Am I allowed to use it, or should I find another space?

Sensors and telemetry

Optional occupancy estimate and door state. Environmental sensors cannot by themselves prove booking permission, a free workstation, a working socket or service availability. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: authoritative booking, check-in and room-use policy.

Analysis required

Permission lookup: treat the authoritative booking and check-in policy as controlling; explain whether any formal release rule applies and direct the student to an authorised alternative when the room remains reserved.

Answer boundary

An empty occupancy estimate or open door never overrides a booking or access rule. If the booking feed is unavailable, the correct answer is that permission cannot be verified.

Why this is useful: A common, realistic boundary case that prevents the chatbot from treating "empty" as "available".

UG-040

CORE

L3 - inclusive group-space recommendation

R2 - integration-dependent

Student question: Which room works best if one group member needs a step-free route and another prefers low noise and low crowding?

Sensors and telemetry

Anonymous room occupancy, corridor footfall, privacy-preserving sound statistics, room air temperature and NDIR CO2 and authorised lift/door status; use time-synchronised sources. No raw audio or speech content may be retained.

Supporting building data

Authoritative: accessible route graph, student-bookable rooms, capacity, equipment and group preferences.

Analysis required

Inclusive forecast and recommendation: treat step-free access, current lift/door status, booking permission and required capacity as hard constraints; forecast crowding and sound for the remaining rooms and explain trade-offs.

Answer boundary

Use preferences only for this request unless the student explicitly saves them. Do not infer disability, health or identity, and do not recommend a room with unverified accessible access.

Why this is useful: Demonstrates inclusive recommendation without treating accessibility as an afterthought.

5. Navigation, accessibility and building services

Questions UG-041-UG-042

UG-041

CORE

L3 - dynamic accessible routing

R2 - integration-dependent

Student question: I need a step-free route to Room 3.02 at 10:55. Which route is likely to be least crowded and least noisy?

Sensors and telemetry

Corridor footfall, privacy-preserving sound-level statistics (LAeq/L10/L90; no raw audio), lift and door state. Use aggregate movement sensing only; lift, door and closure state must come from authorised operational systems.

Supporting building data

Authoritative: authoritative accessible route graph, temporary closures and class-change timetable.

Analysis required

Dynamic accessible-route forecast: remove stairs, restricted paths and unavailable lifts/doors; forecast class-change congestion and sound on the remaining routes; return the fastest acceptable option plus a verified fallback.

Answer boundary

A step-free route must be based on current authoritative accessibility and lift status. A crowding forecast cannot guarantee clear passage; stale status requires an unverified warning and official contact route.

Why this is useful: A technically step-free route may still be difficult when lifts fail or corridors are crowded.

UG-042

CORE

L1 - service status and fallback

R1 - practical with authoritative feed

Student question: Is the lift on my step-free route working now, and what verified alternative should I use if it is not?

Sensors and telemetry

Authoritative lift controller/facilities status and fault timestamp; authorised door status where relevant. Footfall is optional context, not evidence that the lift is working. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: current accessible route graph, verified lift service notices, temporary closures and assistance contact route.

Analysis required

Authoritative current-status lookup: report the lift state, last update and any official outage notice; if unavailable, calculate a verified step-free alternative or state that no independent step-free route is confirmed and provide the assistance contact.

Answer boundary

Do not infer lift operation from movement data or predict that it will remain available. A stale controller or facilities feed means current status is unconfirmed.

Why this is useful: Reduces uncertainty for students who depend on vertical access.

5. Navigation, accessibility and building services

Questions UG-043-UG-044

UG-043

CORE

L3 - sensory-aware routing

R2 - integration-dependent

Student question: Can you route me to my class while avoiding the noisiest and most crowded areas?

Sensors and telemetry

Corridor footfall, room/corridor sound and door/lift status. Use aggregate movement sensing only; lift, door and closure state must come from authorised operational systems. No raw audio or speech content may be retained.

Supporting building data

Authoritative: route graph, destination, access rules and class-change timetable.

Why this is useful: Supports sensory preferences and reduces stress during busy class changes.

Analysis required

Sensory-aware route forecast and recommendation: use a location supplied for this request, compute authorised routes, apply live and short-horizon crowding/sound costs, and explain the extra travel time of the quieter option.

Answer boundary

Use aggregate conditions and avoid continuous tracking. If route, door or lift status is stale, label the result unverified and provide the standard route or official wayfinding source.

UG-044

SITUATIONAL

L2 - entrance recommendation

R2 - verified entrances and counts required

Student question: Which student entrance should I use to reach my next class quickly while avoiding the longest queue?

Sensors and telemetry

Entrance counts, door state and nearby corridor footfall. Use aggregate movement sensing only; lift, door and closure state must come from authorised operational systems.

Supporting building data

Authoritative: verified student entrances, opening/access rules, route graph and destination.

Why this is useful: Useful when several entrances exist or one is temporarily congested or unavailable.

Analysis required

Current lookup and route comparison: remove closed, restricted or non-student entrances; estimate queue delay from doorway counts; combine it with walking time to the destination and return the best authorised option.

Answer boundary

The May 2021 floor plan cannot establish current entrance permissions. Use current access rules and door status; if either is unavailable, provide only verified static directions.

5. Navigation, accessibility and building services

Questions UG-045-UG-046

UG-045

CORE

L2 - waiting-space recommendation

R1 - practical with core feeds

Student question: My appointment is in 30 minutes. Where can I wait nearby in a quiet, uncrowded student-accessible area with reasonable measured conditions?

Sensors and telemetry

Anonymous room occupancy, privacy-preserving sound statistics, air temperature and RH, NDIR CO2, PM2.5 where monitored and illuminance; authorised access/closure status is separate. No raw audio or speech content may be retained.

Supporting building data

Authoritative: appointment location supplied by the user, student-accessible spaces, route graph and opening status.

Analysis required

Current observation plus short forecast: identify nearby public/student waiting candidates, compare sound, estimated occupancy and measured conditions, and check that the option is likely to remain usable for 30 minutes.

Answer boundary

Do not direct the student into staff-only, booked or unsuitable rooms, guarantee a seat or call measured environmental conditions universally comfortable.

Why this is useful: Gives unfamiliar or anxious users a practical place to wait rather than leaving them in circulation areas.

UG-046

CORE

L3 - class-to-class route forecast

R2 - integration-dependent

Student question: I have back-to-back classes on different floors. Which route gives me the best chance of arriving on time?

Sensors and telemetry

Corridor/stair footfall, lift and door state. Use aggregate movement sensing only; lift, door and closure state must come from authorised operational systems.

Supporting building data

Authoritative: the two class locations supplied from the timetable, accessible route graph and class-change patterns.

Analysis required

Time-dependent route forecast: estimate stairs and lift journey time from current footfall and lift state, include expected class-change queues, and return a recommended departure time and backup route.

Answer boundary

Do not promise on-time arrival or encourage unsafe rushing. Use timetable locations only with consent, and label the route unverified when door/lift status is stale.

Why this is useful: Addresses a frequent practical problem in a large multi-floor teaching building.

5. Navigation, accessibility and building services

Questions UG-047-UG-048

UG-047

SITUATIONAL

L2 - privacy-preserving amenity query

R3 - restricted and privacy-sensitive

Student question: Which accessible toilet is nearest and currently open? If privacy-safe occupancy information is available, is another nearby option likely to be less busy?

Sensors and telemetry

No sensing inside toilets. If justified by governance, use only coarse anonymous entrance counts outside the facility and door/closure status; never audio, cameras or identity tracking.

Supporting building data

Authoritative: authoritative accessible-toilet locations, opening/closure state, route graph and facilities notices.

Why this is useful: Supports accessibility and reduces unnecessary movement while preserving privacy.

Analysis required

Privacy-bounded lookup and recommendation: rank verified open accessible toilets by step-free distance; where governance permits, use only coarse recent entrance demand outside the facility and report busyness as low/medium/high with uncertainty.

Answer boundary

No cameras, microphones, identity tracking or occupancy sensing inside toilets. If non-intrusive demand evidence is absent, answer only nearest/open and state that busyness is unknown.

UG-048

CORE

L1 - building-service lookup

R1 - practical with verified asset data

Student question: Where are the nearest water-refill point, vending machine and accessible facilities, and are any reported out of service?

Sensors and telemetry

No environmental sensor is required. Optional authoritative service-status, fault or leak telemetry may indicate temporary unavailability; locations and accessibility come from the current asset/amenity register.

Supporting building data

Authoritative: verified asset and amenity locations, accessibility, opening status and route graph.

Why this is useful: A simple, highly realistic service question that should be answered without technical knowledge.

Analysis required

Verified location and status lookup: return the nearest accessible options, calculate the permitted route and show an outage only when the authoritative service feed is current.

Answer boundary

Do not invent facilities from the May 2021 plan or infer availability from nearby activity. If current asset or outage data are missing, label status unconfirmed.

5. Navigation, accessibility and building services

Questions UG-049-UG-050

UG-049

SITUATIONAL

L3 - proactive route notification

R2 - integration-dependent

Student question: Can you warn me before class if a lift failure, closure or unusual congestion affects my normal route?

Sensors and telemetry

Lift/door status and corridor footfall. Use aggregate movement sensing only; lift, door and closure state must come from authorised operational systems.

Supporting building data

Authoritative: opt-in route or timetable, planned maintenance and closure notices.

Analysis required

Opt-in proactive service: monitor confirmed lift/closure events and statistically unusual route congestion; notify only when the student's chosen route is materially affected, explain whether the cause is confirmed or forecast, and provide a fallback.

Answer boundary

Notifications require explicit opt-in and should not continuously track location. Distinguish authoritative failures/closures from crowding estimates, and allow the student to disable the service.

Why this is useful: Supports proactive planning, especially for students relying on accessible routes.

UG-050

SITUATIONAL

L3 - normal-operation dispersal recommendation

R2 - integration-dependent

Student question: After a large lecture, which normal route out of the teaching area is likely to be less congested?

Sensors and telemetry

Doorway counts, corridor/stair footfall and lift status. Use aggregate movement sensing only; lift, door and closure state must come from authorised operational systems.

Supporting building data

Authoritative: normal permitted route graph, destination direction and temporary closures.

Analysis required

Normal-operations dispersal forecast: estimate post-lecture flows from doorway and corridor counts, rank permitted routes by expected delay and, where necessary, distribute recommendations to avoid creating a new bottleneck.

Answer boundary

This applies only after normal class dismissal. During any alarm or staff-directed incident, official evacuation instructions override the recommendation; lifts must not be recommended for fire evacuation.

Why this is useful: Reduces avoidable bottlenecks after large classes.

6. Indoor environmental quality and well-being

Questions UG-051-UG-052

UG-051

SITUATIONAL

L4 - diagnostic explanation

R3 - research-grade

Student question: This room feels stuffy even though the CO2 reading is normal. What else could be contributing?

Sensors and telemetry

NDIR CO2 sensor health and placement, temperature, humidity, indicative TVOC, PM2.5/PM10, airflow and occupancy. Use room-level, commissioned sensors with health metadata and outdoor/reference conditions where relevant. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: recent cleaning/activity reports and room/HVAC relationships where available.

Analysis required

Diagnostic assessment: first verify CO2 sensor placement and health; then examine temperature, RH, airflow, PM, indicative TVOC, occupancy and recent activity; separate direct observations from plausible explanations and rank them by support.

Answer boundary

A generic TVOC sensor cannot identify a chemical, and the system cannot diagnose a health condition. Normal CO2 does not disprove a complaint; incomplete room sensing requires a limited answer and official reporting option.

Why this is useful: Prevents the building from reducing every air-quality complaint to CO2 alone.

UG-052

SITUATIONAL

L4 - matched spatial comparison

R3 - research-grade

Student question: Has this study area really been warmer than similar spaces once occupancy, weather and time of day are taken into account?

Sensors and telemetry

Room air temperature plus globe/radiant-temperature reference where available, anonymous occupancy, outdoor temperature and solar radiation; use commissioned sensors and comparable spatial coverage.

Supporting building data

Authoritative: room type, size, facade orientation and HVAC zone.

Analysis required

Matched spatial comparison: select rooms with comparable use, size, facade exposure and HVAC service; adjust for occupancy, time and outdoor conditions; report the excess temperature distribution, recurrence and uncertainty.

Answer boundary

Do not compare unlike spaces or infer a fault from a higher raw average. If radiant conditions or relevant zone coverage are absent, describe only the measured air-temperature difference.

Why this is useful: Helps distinguish a persistent location-specific issue from normal variation.

6. Indoor environmental quality and well-being

Questions UG-053-UG-054

UG-053

CORE

L2 - stability recommendation

R1 - practical with core feeds

Student question: Which student-accessible study area has had the most stable temperature and lighting so far today?

Sensors and telemetry

Room air temperature and illuminance time series for the requested day, plus anonymous occupancy and sensor-health metadata; compare only verified student-accessible spaces.

Supporting building data

Authoritative: student-accessible spaces, current availability and route graph.

Analysis required

Current-day stability comparison: calculate temperature and illuminance variability over an appropriate recent window, check trend and sensor health, and rank only verified available spaces; show that "stable" is separate from preferred level.

Answer boundary

Stable conditions are not automatically comfortable or suitable. Ask for preferred ranges when needed, and do not rank spaces with missing or poorly placed sensors.

Why this is useful: Useful for long study periods and visually sensitive work.

UG-054

CORE

L2 - historical acoustic comparison

R1 - practical with sound statistics

Student question: Which student-accessible study spaces have had the lowest and most stable background sound over the last four weeks?

Sensors and telemetry

Room-level privacy-preserving LAeq/L10/L90 sound statistics and anonymous occupancy, with verified placement, uptime and opening-hour coverage. No raw audio or speech content.

Supporting building data

Authoritative: space-use rules, timetable and opening hours.

Analysis required

Historical acoustic comparison: compare median, upper-percentile and variability by equivalent times of day; exclude closed/unused periods; rank spaces by both typical quietness and reliability over four weeks.

Answer boundary

Low average sound does not prove conversational privacy or current seat availability. Unmonitored or intermittently monitored spaces must be labelled unknown.

Why this is useful: Provides a robust quiet-study recommendation based on repeated behaviour.

6. Indoor environmental quality and well-being

Questions UG-055-UG-056

UG-055

SITUATIONAL

L4 - noise-source investigation

R3 - research-grade

Student question: The noise here is distracting. Does it follow class-change times, plant operation or a one-off event?

Sensors and telemetry

Privacy-preserving sound-level statistics (LAeq, L10 and L90; no raw audio); authorised plant run-state and vibration only where purposefully instrumented; anonymous occupancy and corridor footfall.

Supporting building data

Authoritative: timetable, maintenance events and equipment-to-space relationships.

Analysis required

Noise-source diagnostic assessment: detect recurrence by time and day, align events with class changes and authorised plant run-state/vibration, compare nearby zones, and rank plant, movement and one-off explanations by support.

Answer boundary

Do not record or identify speakers. Timing correlation does not prove a mechanical cause; if plant telemetry is absent, report only the observed recurrence pattern and recommend inspection.

Why this is useful: Supports a better fault report and avoids treating all noise events as the same problem.

UG-056

SITUATIONAL

L3 - glare diagnosis and recommendation

R3 - validated daylight model required

Student question: Is the brightness here mainly daylight, electric lighting or both, and where in the room is glare likely to be lower?

Sensors and telemetry

Task-plane and vertical illuminance, outdoor solar irradiance, lighting circuit/state, blind position and room-zone temperature; use validated geometry or a privacy-preserving luminance method that stores no identifiable imagery.

Supporting building data

Authoritative: room orientation, window and luminaire layout, current weather and accessible seating zones.

Analysis required

Diagnostic assessment plus recommendation: estimate daylight and electric-light contributions, calculate zone-level glare risk with a validated model, and recommend a lower-glare authorised area with confidence.

Answer boundary

Illuminance alone is not a complete glare measure. If window/luminaire geometry or suitably placed light data are absent, report brightness and lighting state only, not cause or glare comfort.

Why this is useful: Helps students with visual discomfort choose a better location and supports lighting diagnosis.

6. Indoor environmental quality and well-being

Questions UG-057-UG-058

UG-057

CORE

L3 - low-stimulation recommendation

R2 - integration-dependent

Student question: Which available area is likely to have lower noise, lower crowding, less glare and a more stable temperature for the next hour?

Sensors and telemetry

Privacy-preserving sound statistics, anonymous room occupancy, corridor footfall, illuminance, air temperature and solar-radiation context; use room-level, time-synchronised data. No raw audio or speech content may be retained.

Supporting building data

Authoritative: student-accessible spaces, route graph and upcoming timetable activity.

Analysis required

Multi-criteria short-horizon forecast: keep noise, crowding, glare and temperature as separate criteria; forecast each for one hour, apply any accessibility/access constraints, and explain the trade-offs behind the ranked options.

Answer boundary

Do not infer a disability or health condition from the request. Save preferences only with consent, and omit any criterion whose sensing or geometry is insufficient.

Why this is useful: Provides a practical low-stimulation option without requiring the student to disclose a diagnosis.

UG-058

SITUATIONAL

L3 - post-cleaning background check

R2 - additional sensing and records required

Student question: This room smells strongly after cleaning. Are the monitored conditions back near the room's usual background, and should I report the smell?

Sensors and telemetry

Indicative TVOC, PM2.5/PM10, NDIR CO2, temperature, RH and ventilation airflow, all with room-level placement, recent health status and an outdoor/reference context where relevant.

Supporting building data

Authoritative: authoritative cleaning time and broad product/process category where available, room baseline, ventilation/maintenance status and the official reporting route.

Analysis required

Event recovery assessment: use the verified cleaning time, compare TVOC/PM and ventilation trends with the room's normal background, check sensor agreement, and advise reporting when odour, abnormal readings or symptoms persist.

Answer boundary

A generic TVOC sensor cannot identify a substance or declare the room safe. A strong odour or symptoms warrant leaving the area and using the official reporting route even when sensors appear normal.

Why this is useful: Gives students a bounded way to respond to an unusual odour without overclaiming from low-cost sensors.

6. Indoor environmental quality and well-being

Questions UG-059-UG-060

UG-059

SITUATIONAL

L4 - indoor/outdoor source comparison

R3 - research-grade

Student question: Particle levels are higher today. Is the pattern more consistent with outdoor air entering the building or with activity inside?

Sensors and telemetry

Indoor and outdoor PM2.5/PM10, authorised window/door and ventilation state, anonymous occupancy, and relevant aggregate activity/equipment signals; include humidity and sensor-health context.

Analysis required

Indoor/outdoor source assessment: compare outdoor and indoor PM timing, magnitude and lag; examine ventilation, door/window state and scheduled indoor activity; rank source hypotheses and show uncertainty rather than naming a proven source.

Supporting building data

Authoritative: weather, cleaning/workshop schedule and building airflow relationships.

Answer boundary

Optical particle counts cannot identify chemical composition or prove origin. Poor calibration, humidity artefacts or missing airflow context require a direct-observation-only answer.

Why this is useful: Explains an air-quality warning more usefully than merely reporting a high number.

UG-060

CORE

L4 - answerability assessment

R2 - evidence metadata required

Student question: The building says this room is comfortable. What did it measure, what did it not measure, and how confident should I be?

Sensors and telemetry

Available temperature, humidity, radiant-temperature, air-speed and occupancy measurements; sensor health. Use room-level, commissioned sensors with health metadata and outdoor/reference conditions where relevant. Occupancy and footfall must remain aggregate and non-identifying.

Analysis required

Claim-to-evidence assessment: map the comfort statement to air temperature, mean radiant temperature, air speed, RH, activity, clothing and user feedback; show which factors are measured, assumed or missing and classify the claim as fully, partly or not assessable.

Supporting building data

Authoritative: the stated comfort criterion, room use and any user feedback.

Answer boundary

Temperature and RH alone cannot establish comfort for every person. The system must not fill missing personal factors with hidden assumptions or convert a partial environmental check into a universal comfort claim.

Why this is useful: Builds evidence literacy and prevents confident claims from incomplete sensing.

7. Sustainability and responsible building use

Questions UG-061-UG-062

UG-061

SITUATIONAL

L3 - low-impact space recommendation

R3 - advanced energy integration

Student question: If two study areas meet my needs, which one is already open and in use, so the building is less likely to serve another nearly empty area?

Sensors and telemetry

Anonymous occupancy, room temperature and RH, NDIR CO2, privacy-preserving sound statistics, lighting/HVAC operating state and zone energy where the meter boundary is defined. No raw audio or speech content may be retained.

Supporting building data

Authoritative: student-accessible/open spaces, bookings, room-use rules, route graph and operational policy on opening or conditioning spaces.

Analysis required

Low-impact recommendation: first satisfy student access, accessibility, quietness and measured-condition requirements; then prefer an already-open, conditioned space with spare capacity and estimate incremental energy only where control and meter boundaries support it.

Answer boundary

Do not pressure students into crowded, inaccessible or poor-quality spaces or claim that one student's choice will close a room. If incremental energy cannot be estimated, omit the energy ranking.

Why this is useful: Supports a sustainability-conscious choice without shifting building-management responsibility onto students.

UG-062

CORE

L2 - waste-reporting query

R2 - BMS and occupancy integration

Student question: This room looks empty, but the lights and ventilation are on. Is that expected, or should I report it?

Sensors and telemetry

Independent occupancy, lighting state/illuminance, HVAC airflow/valve state and energy. Use meters with documented boundaries and the relevant operating-state telemetry. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: bookings, preconditioning schedule, after-hours policy and reporting route.

Analysis required

Observation plus anomaly assessment: verify likely vacancy using independent sources; check bookings, preconditioning, cleaning and safety operation; determine whether lighting/HVAC operation is persistent and unusual, then prepare a concise report.

Answer boundary

Occupancy sensors can miss people, and legitimate preconditioning or safety operation is not waste. Without clear meter/control boundaries and operational context, report the observation but do not label it avoidable energy.

Why this is useful: Turns a common student observation into a useful facilities report while reducing false alarms.

7. Sustainability and responsible building use

Questions UG-063-UG-064

UG-063

SITUATIONAL

L2 - room-energy comparison

R2 - submetering required

Student question: For a student sustainability project, how much electricity does this teaching room typically use during a class compared with similar rooms under similar conditions?

Sensors and telemetry

Room/zone electricity and HVAC thermal/electrical meters with documented service boundaries, anonymous occupancy, air temperature and NDIR CO2 and relevant operating-state telemetry.

Supporting building data

Authoritative: room size, class duration, attendance, timetable and weather.

Analysis required

Matched energy comparison: select sessions with similar room type, duration, attendance and weather; separate base load from class-associated load where meter boundaries permit; normalise transparently and return uncertainty.

Answer boundary

Do not attribute all zone or HVAC energy to one class when meters serve shared assets. If boundaries or attendance data are inadequate, provide only the measured aggregate and mark comparison not assessable.

Why this is useful: Makes building-energy information understandable without unfairly assigning responsibility to one student group.

UG-064

SITUATIONAL

L3 - right-sized room recommendation

R2 - booking integration required

Student question: Can you suggest a smaller student-bookable room for four people that still meets our accessibility, power, air-quality and acoustic needs?

Sensors and telemetry

Anonymous occupancy, room air temperature, direct NDIR CO2 and LAeq/L10/L90 sound statistics; current electrical-service status where available. Capacity, sockets and accessibility are authoritative room attributes, not sensor inferences. No raw audio or speech content may be retained.

Supporting building data

Authoritative: student-bookable rooms, capacity, access, features and bookings.

Analysis required

Right-sizing recommendation: treat student bookability, step-free access, capacity and power provision as hard constraints; forecast occupancy and measured conditions, then rank the smallest suitable authorised rooms and explain trade-offs.

Answer boundary

Right-sizing must not reduce accessibility, privacy or room quality. Do not recommend a non-bookable room or claim power/socket availability from incomplete asset or fault data.

Why this is useful: Supports efficient room use while keeping the group's practical needs central.

7. Sustainability and responsible building use

Questions UG-065-UG-066

UG-065

EXPLORATORY

L4 - conditional behaviour-impact analysis

R3 - research-grade and policy-dependent

Student question: For a sustainability project, when students are allowed to open windows, what usually happens to temperature, CO2, particles and energy use?

Sensors and telemetry

Window state, indoor environmental data, HVAC state/energy and outdoor temperature, wind and PM2.5/PM10. Use meters with documented boundaries and the relevant operating-state telemetry.

Supporting building data

Authoritative: window-use policy, room/HVAC relationships and occupancy.

Analysis required

Conditional intervention analysis: compare matched open- and closed-window periods under similar weather, occupancy and HVAC operation; estimate changes in temperature, CO2, PM and energy, including lags and uncertainty.

Answer boundary

Do not recommend opening a window when policy, security, weather or outdoor pollution makes it inappropriate. Association is not causal proof unless confounding is adequately controlled and meter boundaries are valid.

Why this is useful: Provides evidence about a common occupant action and its environmental trade-offs.

UG-066

SITUATIONAL

L2 - daylight recommendation

R3 - validated daylight and lighting integration required

Student question: Which student study areas currently have enough daylight for reading without excessive glare or unnecessary electric lighting?

Sensors and telemetry

Task-plane and vertical illuminance, outdoor solar irradiance, lighting state/energy, blind position and occupancy by candidate zone; use verified placement and a validated glare/daylight model. Occupancy and footfall must remain aggregate and non-identifying.

Supporting building data

Authoritative: room orientation, window/luminaire layout, access and availability.

Analysis required

Daylight and lighting recommendation: identify zones with sufficient task light, estimate glare risk, check whether electric lighting is on and rank authorised spaces without simply choosing the brightest area.

Answer boundary

Daylight adequacy depends on task and user. Without validated glare geometry or lighting state, report only measured illuminance; do not claim avoidable lighting energy without a defined meter/control boundary.

Why this is useful: Links student visual comfort with responsible use of lighting.

7. Sustainability and responsible building use

Questions UG-067-UG-068

UG-067

SITUATIONAL

L4 - intervention evaluation

R3 - research-grade

Student question: The building recently changed its controls to save energy. Did it reduce consumption without making teaching or study conditions worse?

Sensors and telemetry

Electricity/thermal-energy meters with defined boundaries; air temperature and RH, NDIR CO2, supply airflow, anonymous occupancy and privacy-preserving sound statistics where relevant. No raw audio or speech content may be retained.

Supporting building data

Authoritative: intervention date, timetable, weather, complaints and control configuration.

Analysis required

Intervention evaluation: use occupancy- and weather-adjusted pre/post or interrupted time-series analysis; assess energy, temperature, CO2, airflow, sound and complaints separately; test persistence and alternative explanations.

Answer boundary

A simple before/after average does not establish causality. Concurrent timetable, weather, maintenance or sensor changes must be disclosed, and shared meter boundaries may limit attribution.

Why this is useful: Allows students and representatives to scrutinise whether sustainability measures protect learning conditions.

UG-068

EXPLORATORY

L4 - after-hours waste analysis

R3 - restricted operational data

Student question: For our student sustainability project, which floors appear to use the most energy after normal teaching hours without clear occupancy or operational need?

Sensors and telemetry

Floor or zone electricity and thermal-energy meters with documented boundaries; anonymous occupancy; lighting and HVAC operating states.

Supporting building data

Authoritative: opening hours, events, cleaning, maintenance, preconditioning schedules, and known server, safety and other essential loads.

Analysis required

Aggregated after-hours assessment: define normal teaching hours and legitimate operation, combine floor/zone meters with occupancy and control states, estimate residual potentially avoidable load and uncertainty, and separate essential/base loads.

Answer boundary

Student-facing results must be aggregated and must not expose security-sensitive plant operation. Do not label energy avoidable when meter boundaries, occupancy or operational schedules are incomplete.

Why this is useful: Supports student sustainability engagement while preserving operational context.

7. Sustainability and responsible building use

Questions UG-069-UG-070

UG-069

EXPLORATORY

L4 - multi-objective recommendation

R3 - research-grade

Student question: Can the building suggest study times and spaces that meet my needs while avoiding extra energy use from opening or conditioning a separate area?

Sensors and telemetry

Anonymous occupancy, room temperature and NDIR CO2 and sound forecasts, lighting/HVAC operating state, and energy meters with documented boundaries; no personal-device tracking. No raw audio or speech content may be retained.

Supporting building data

Authoritative: student preferences, available spaces, timetable, bookings and route graph.

Analysis required

Multi-objective recommendation: first satisfy access, accessibility and study-quality constraints; then estimate whether choosing an already-open or better-utilised space could reduce incremental operation; expose trade-offs and let the student reject the option.

Answer boundary

Sustainability must not be used to pressure students or shift facilities responsibility to them. Without valid control/meter relationships, omit the energy claim and recommend on student criteria only.

Why this is useful: Explores how environmental and student goals can be considered together rather than treated as opposites.

UG-070

CORE

L4 - evidence transparency

R2 - evidence metadata required

Student question: When the building makes a sustainability claim, what measurements support it, and what important areas are not monitored?

Sensors and telemetry

For each statement, the required energy/water meters, anonymous occupancy and environmental sensors, with location, meter boundary, calibration/commissioning, uptime, sampling and spatial/temporal coverage metadata.

Supporting building data

Authoritative: authoritative wording of each sustainability statement, target or baseline, reporting period, intervention record and method used for institutional reporting.

Analysis required

Statement-to-evidence assessment: decompose each sustainability statement into variable, place, period, baseline and method; verify source authority, meter/sensor boundaries, calibration and coverage; classify supported, partly supported, conflicting or not assessable.

Answer boundary

Do not convert promotional language into a technical conclusion without a defined target and auditable method. Unmonitored areas, changed baselines and missing evidence must remain explicit.

Why this is useful: Gives students a critical, evidence-aware view of institutional sustainability communication.

8. Faults, disruptions, trust and safety

Questions UG-071-UG-072

UG-071

SITUATIONAL

L4 - fault-scope diagnosis

R3 - research-grade

Student question: Is this room's unusual condition local, shared by nearby rooms, or part of a wider floor problem?

Sensors and telemetry

Room, adjacent-room and corridor environmental measurements; HVAC telemetry and occupancy; outdoor weather. Use time-synchronised observations and authorised event/status feeds with provenance. Occupancy and footfall must remain aggregate and non-identifying.

Analysis required

Spatial diagnostic assessment: compare the room, adjacent rooms and floor references with common timestamps; adjust for occupancy and facade effects; trace shared HVAC relationships and classify the issue as likely local, shared-zone or wider-floor with confidence.

Supporting building data

Authoritative: room adjacency, facade context and equipment-to-space relationships.

Answer boundary

Do not name a failed asset from spatial similarity alone. If adjacent rooms are unmonitored or service relationships are incomplete, report the observed scope and direct the student to the official reporting route.

Why this is useful: Helps a student submit a more useful report to the correct operational team.

UG-072

CORE

L3 - report-or-wait recommendation

R2 - integration-dependent

Student question: The room feels unusually cold. Is this brief and expected, or persistent enough that I should report it?

Sensors and telemetry

Room temperature, radiant-temperature reference where available, HVAC state and neighbouring-room conditions. Use time-synchronised observations and authorised event/status feeds with provenance.

Analysis required

Report-or-wait recommendation: compare the current episode with the room's normal behaviour under similar weather and occupancy; assess duration, magnitude and neighbouring-room pattern; recommend reporting when persistence or severity exceeds a stated rule.

Supporting building data

Authoritative: outdoor weather, room baseline, current duration and reporting route.

Answer boundary

Do not dismiss subjective feedback because a single sensor is within a broad range, and do not diagnose a fault without operational evidence. Show missing radiant/airflow information where relevant.

Why this is useful: Reduces unnecessary reports while encouraging prompt escalation of persistent problems.

8. Faults, disruptions, trust and safety

Questions UG-073-UG-074

UG-073

CORE

L1 - issue-status query

R2 - maintenance integration required

Student question: Has this room or equipment problem already been reported, and what status can students see?

Sensors and telemetry

Relevant room anomaly or equipment status, if available. Use time-synchronised observations and authorised event/status feeds with provenance.

Analysis required

Authoritative issue-status lookup: match location and asset to an existing ticket, show only student-visible states such as received, assigned, planned or resolved, and provide the official reporting route when no reliable match exists.

Supporting building data

Authoritative: maintenance ticket matching by location/asset and an authorised public-status field.

Answer boundary

Do not expose reporter identity, staff notes, access/security details or unrestricted maintenance records. If the ticket system cannot be queried, say status is unavailable rather than inferring it from sensors.

Why this is useful: Prevents duplicate reports and reassures students that a known issue is being handled.

UG-074

SITUATIONAL

L4 - repair verification

R3 - research-grade

Student question: Facilities worked on this room yesterday. Did the measured conditions actually improve afterwards?

Sensors and telemetry

The environmental or equipment measurements relevant to the fault. Use time-synchronised observations and authorised event/status feeds with provenance.

Analysis required

Repair-effect evaluation: use the work-order timestamp and fault-specific measurements, compare matched pre/post periods or an interrupted time series, adjust for occupancy, timetable and weather, and test whether the change persisted.

Supporting building data

Authoritative: intervention time, work-order scope, occupancy, timetable and weather.

Answer boundary

One post-repair reading is not evidence of success. Do not disclose restricted work-order details or claim causality when operating conditions changed materially.

Why this is useful: Makes repair outcomes transparent and supports follow-up when a problem persists.

8. Faults, disruptions, trust and safety

Questions UG-075-UG-076

UG-075

CORE

L2 - pre-class service check

R2 - multi-system integration required

Student question: Before I walk to class, are the assigned room, network, audiovisual equipment, power and monitored environmental systems reported as working normally?

Sensors and telemetry

Room air temperature, RH and NDIR CO2; monitored electrical distribution and equipment state where available. Network and audiovisual status must come from their authoritative management systems, not be inferred from environmental or power data.

Supporting building data

Authoritative: timetable/room assignment, network and audiovisual service status, planned maintenance, current fault notices and room access state.

Analysis required

Multi-system readiness lookup: check the assigned room and each service independently; label confirmed normal, confirmed fault, stale, unmonitored or unknown; give the official support route and, only if authorised, a verified alternative.

Answer boundary

Missing telemetry is not normal operation. Do not guarantee every socket, user device, network session or piece of AV equipment; report only monitored authoritative status and timestamps.

Why this is useful: Avoids wasted journeys and late discovery of a teaching-room or service failure.

UG-076

CORE

L2 - disruption-aware routing

R2 - authoritative incident feed required

Student question: A leak has been reported near my usual study area. Which permitted route and alternative space should I use?

Sensors and telemetry

Authorised leak/incident event and area/door status; aggregate occupancy and environmental conditions only for candidate alternatives.

Supporting building data

Authoritative: official closure boundary and notice, current route graph, student-accessible alternative spaces and facilities/safety authority timestamp.

Analysis required

Incident-aware route and space recommendation: use the official affected boundary, exclude that zone, calculate permitted routes and rank verified alternative study spaces by distance and current measured conditions.

Answer boundary

A low-cost leak sensor must not define the hazard area. If the official closure boundary or route status is unavailable, direct the student to facilities/security and do not invent a safe route.

Why this is useful: Combines incident information with a practical next action.

8. Faults, disruptions, trust and safety

Questions UG-077-UG-078

UG-077

CORE

L1 - temporary-closure route query

R1 - practical with closure feed

Student question: During a planned or temporary disruption, which normal entrances, lifts and routes are currently verified as usable?

Sensors and telemetry

Authoritative door and lift status; optional footfall. Use time-synchronised observations and authorised event/status feeds with provenance. Occupancy and footfall must remain aggregate and non-identifying.

Analysis required

Verified disruption lookup: apply current official closures and access rules to the route graph, report usable normal entrances/lifts/routes with timestamps and provide alternatives for unavailable links.

Supporting building data

Authoritative: official closures, access rules and route graph.

Answer boundary

This applies to planned or temporary normal-operation disruption, not an active emergency. During an alarm, official instructions and staff direction take precedence; stale status must be labelled unverified.

Why this is useful: Useful during maintenance or local closures while preserving the safety hierarchy.

UG-078

CORE

L4 - conflicting-evidence query

R2 - quality metadata required

Student question: Two nearby sensors disagree. Which reading is more reliable, and can the building still answer my question?

Sensors and telemetry

Raw readings plus sensor location, height, calibration/commissioning, last update, drift/health indicators and independent nearby references where physically comparable.

Analysis required

Conflicting-evidence assessment: check units, timestamps, placement, calibration/commissioning and sensor health; determine whether the sensors represent different micro-zones; compare independent references; avoid automatic averaging and return supported, conflicting or not assessable.

Supporting building data

Authoritative: room/zone boundaries, sensor maintenance and relocation history, units, sampling interval and the exact decision the reading is intended to support.

Answer boundary

Do not select the convenient reading. If unresolved disagreement could change the answer, show the conflict and withhold the conclusion rather than creating false precision.

Why this is useful: Makes data conflict visible and prevents false certainty.

8. Faults, disruptions, trust and safety

Questions UG-079-UG-080

UG-079

CORE

L2 - provenance query

R1 - provenance by design

Student question: Why should I trust this answer, and can you show me the sources used, how recent they are and what evidence is missing?

Sensors and telemetry

Every sensor and system source actually used, with location, timestamp, health, calibration/commissioning and coverage metadata.

Supporting building data

Authoritative: the executed query/analysis record, spatial and temporal scope, authoritative contextual sources, permissions, transformations and material excluded evidence.

Analysis required

Provenance explanation: list the evidence actually used and its spatial/time scope; separate observed, looked-up, calculated, inferred, forecast and recommended statements; show freshness, missingness, confidence and why candidate alternatives were excluded.

Answer boundary

Trust must come from inspectable evidence and honest limits, not fluent wording. Apply the student's permissions and summarise security-sensitive operational detail rather than exposing it.

Why this is useful: Lets students inspect the evidence and judge whether a recommendation is well supported.

UG-080

CORE BOUNDARY CASE

L1 - safety-bounded incident response

R3 - restricted authoritative integration

Student question: The fire alarm is sounding. What should I do?

Sensors and telemetry

No experimental sensor is required for the immediate instruction. Authorised status from certified alarm systems may be shown only if it is current and does not delay action.

Supporting building data

Authoritative: official alarm, signage, staff/fire-warden instructions, University emergency procedures and emergency contact information.

Analysis required

Immediate safety instruction, not analytics: tell the student to leave promptly by the nearest safe signed exit, not use lifts, follow staff or fire-warden instructions, and go to the designated assembly point shown by official signage or instructions. Contact emergency services or University Security when instructed or necessary.

Answer boundary

The interface must not wait for sensor analysis, diagnose whether the alarm is genuine, calculate a personalised evacuation route, delay departure or replace certified alarms, signage, staff, fire wardens or emergency responders.

Why this is useful: Provides immediate, authoritative action while keeping the chatbot subordinate to certified alarms and human instructions.

4. Catalogue summary and implementation requirements

The catalogue combines practical status questions with integration-dependent forecasts, multi-source assessments and restricted safety cases. Readiness identifies the verified evidence and controls required for a responsible answer.

Questions	80	Eight student-centred sections containing ten questions each.
R1 practical	12	Can be supported with the proposed core sensing and basic authoritative feeds.
R2 integration-dependent	45	Require additional timetable, booking, BMS, IT, lift, maintenance or forecasting integration.
R3 research/restricted	23	Require dense sensing, validated reasoning, controlled data, or stronger privacy and safety governance.

Recurring system capabilities

- 1

Space and route reasoning
Resolve rooms, floors, student access, route constraints, proximity and temporary closures.
- 2

Environmental and occupancy evidence
Use correctly located room sensors, anonymous occupancy and corridor footfall without overclaiming seat-level certainty.
- 3

Forecasting and recommendation
Predict crowding, sound and room conditions, rank alternatives and expose confidence and fallback options.
- 4

Institutional integration
Combine timetables, bookings, room features, network or AV status, maintenance, lift status and operational feeds.
- 5

Evidence quality and provenance
Check calibration, sensor health, missingness, spatial coverage and conflicts before answering.
- 6

Clarification and abstention
Ask for missing constraints, refuse unauthorised requests and return not assessable when evidence is inadequate.

QUESTION RECORD STRUCTURE

Each catalogue entry can be converted into a system test containing the user utterance and paraphrases, required entities, minimum evidence, expected analysis, authorisation rule, answerability class, acceptable clarification, expected answer characteristics and prohibited overclaims.

5. Evidence, permissions and answerability rules

These rules apply across all 80 records and define when the building may answer, qualify, ask for clarification or return not assessable.

Building and room status

The May 2021 plan is a historical spatial reference. Current room purpose, capacity, student access, bookability, entrances, amenities, lifts, services and closures require an authoritative current source.

Spatial coverage

A room-level answer requires a correctly located sensor inside that room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

Time window and freshness

Every answer must resolve the requested time period, align source timestamps and show the latest update. Stale observations or feeds are not current status.

Calibration and sensor health

Use commissioning/calibration records, uptime, missingness, drift and plausibility checks. A failed or unverified sensor cannot support a precise comparison, forecast or diagnosis.

Occupancy, seats and privacy

Use anonymous aggregate counts. Occupancy estimates do not identify people or guarantee a seat, workstation or room availability; booking and access remain authoritative.

Sound and recording

Routine acoustic services use numerical LAeq/percentile statistics. Do not retain speech, infer conversation content or claim acoustic privacy from a low sound level alone.

Environmental interpretation

NDIR CO2 is a ventilation indicator, not a complete air-quality verdict. Temperature, RH, PM, indicative TVOC, airflow, outdoor conditions and activity may be relevant, subject to sensor quality.

Comfort and personal variation

Temperature alone does not establish comfort. Radiant conditions, air movement, humidity, clothing, activity and personal preference may be missing; state partial answerability.

Operational services and permissions

Bookings, access, lifts, network, AV, maintenance and closures come from authorised systems. Missing telemetry is unknown, not normal; enforce the student role and minimise personal data.

Safety, uncertainty and refusal

Certified alarms, official communications and human instructions take precedence. Forecasts need confidence and refresh times; missing or conflicting critical evidence requires clarification, a limited answer or not assessable.

FINAL ANSWERABILITY RULE

Answer only when the required variables, current spatial coverage, time window, source authority, data quality and permissions are adequate. Otherwise ask for clarification, omit unsupported criteria, provide a limited answer with explicit uncertainty, or return not assessable.